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Improving Forecast Accuracy as an Effort to Enhance Business Performance

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ABSTRACT

This research investigates the role of forecasting accuracy in improving business performance. Forecasting has become a critical managerial tool in dynamic business environments where demand fluctuations and market uncertainty often challenge strategic decision-making. The main objective of this research is to empirically test whether higher forecasting accuracy contributes significantly to enhancing business performance. A quantitative approach was employed using survey data collected from firms in hospitality and the food and beverage industries. Data were analyzed using Structural Equation Modeling – Partial Least Squares (SEM-PLS) to examine the relationships among data quality, forecasting techniques, forecasting accuracy, and business performance. The findings indicate that forecasting accuracy mediates the effect of data quality and forecasting methods on business performance. Furthermore, the study concludes that big data capabilities and structured budgeting significantly improve forecast accuracy, with big data being the most influential factor, while competitive intensity does not exert a significant direct impact. Importantly, forecast accuracy itself strongly and positively affects business performance, underscoring its role as a critical mediator between organizational capabilities and performance outcomes. The study contributes to the literature by providing empirical evidence on the mediating role of forecasting accuracy and offers practical implications for managers to design more accurate forecasting systems that support sustainable business growth.

KEYWORDS *forecasting accuracy; business performance.*

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INTRODUCTION

In an increasingly complex and competitive business era, a company's ability to make the right and fast decisions is a key factor in maintaining competitiveness. One of the important elements that supports the strategic decision-making process is the accuracy in forecasting. Accurate forecasting allows companies to forecast market demand, plan inventory needs, set production schedules, and manage resources efficiently. In practice, however, many organizations still face challenges in producing accurate predictions due to limited historical data, the analytical capabilities of relevant stakeholders, conventional forecasting methods that rely on traditional intuition-based forecasting, and the lack of integration of advanced technology in the planning process. The phenomenon gap from previous research reveals that While strategic decision-making plays an important role, many companies face major challenges in this process, especially if they are not supported by predictive analytics (Adesina et al., 2024). One of the main challenges is the reliance on incomplete or inaccurate data (Balakrishnan et al., 2025). Conventional methods tend to be inadequate in responding to complex and dynamic data characteristics (Olamijuwon, 2024). In many cases, decision-makers must rely on their historical data or intuition, which may not fully reflect the complexity of the current business environment. This can lead to decisions made based on outdated or irrelevant information, increasing the risk of suboptimal decision-making. The limitation of the problem in this study focuses on the relationship between forecast accuracy with business performance. This research is limited to companies in the hospitality industry in the Greater

Jakarta area assuming that respondents have access to and understanding of the forecasting system and historical data related to actual forecasting. The unit of analysis in this study is the company level, not the individual.

Forecasting is the foundation revenue management and hotel operations, accurate prediction allows optimal rate determination, room allocation, supply chain management, scheduling Staff so that it directly affects the profitability of the company. There are traditional methods and machine learning in forecasting for future business projections. Traditional methods are still widely used today, but they are limited when dealing with the complex influence of external influences of a company. By utilizing machine learning & deep learning such as channel / online travel agent data, event calendar, competitor prices and climate, then the forecasting function will be more optimal. Technique budgeting involves the processing of numbers, estimation, attention to detail, application forecasting in an effort to provide the right strategic decisions (Homauni et al., 2023). The use of variables such as big data to improve forecasting, can allow companies to maintain cost efficiency and affect overall performance (Wellens et al., 2023). Through the use of large volumes of data, advanced analytics, companies can gain a deeper understanding of business operation, predict future trends and respond quickly to changing market conditions (Paramesha et al., 2024). Every business needs a specific strategy to maintain the sustainability of the company in an increasingly competitive business environment (Yuniawati & Farman, 2023).

The contribution of this research is intended for business practitioners as a reference, especially operational managers and those related in developing a more accurate and adaptive forecasting system. By understanding the factors that affect forecasting accuracy, companies can adopt more appropriate strategies and technologies to minimize operational risk and optimize business decision-making. This research makes a theoretical contribution by confirming that forecasting accuracy is not only a technical activity in operations management, but also has strategic implications for improving business performance. In addition, this study also contributes to business performance theory by showing that the accuracy of forecasting is closely related to the achievement of performance indicators such as profitability, customer satisfaction, and operational effectiveness. Thus, this research not only strengthens existing theories, but also presents a conceptual framework that integrates aspects of forecasting, operational efficiency, and business performance, thus opening up space for more comprehensive theory development.

Previous studies have shown that improving the accuracy of forecasting still faces a number of challenges, including unpredictable market dynamics (Balakrishnan et al., 2025), the limitations of traditional forecasting methods that are less adaptive to changing demand trends, as well as the constraints of data integration across departments within the organization. There are not many integrated system providers that comprehensively combine forecasting functions in a company's system (Siddiqui et al., 2022). What is the economic impact that depends on the level of competitiveness, especially in the tourism sector (Fernandez, 2022). Departing from limitations to reliability financial forecasting as well as future research and risk analysis, this study proposes novelty by examining integratively the relationship between forecasting accuracy and business performance, by including variables such as budgeting, Big Data and competitiveness as an independent variable and forecasting accuracy as a mediation variable. Thus, this approach will provide a more comprehensive conceptual framework, particularly in the industrial context hospitality such as hospitality which is heavily influenced by market uncertainty. Seasonal patterns in tourism businesses are considered a major concern and are considered quite a complex variable, but strategic in reading competitiveness can maximize the success of a tourism business performance by understanding accurate business forecasting strategies (Yabanci, 2023).

This research provides information on quality data, understanding trends, systems budgeting and technology Big Data It is seen as a strategic resource that, if managed properly, will improve the accuracy of forecasting. (Salamah, 2023) concludes that effective financial management is influenced by various factors such as foreasting, budgeting & planning, has a crucial role in driving business performance improvement. (Adesina et al., 2024) Finding that predictive analytics has a profound impact on business performance through the provision of accurate data. (Eton et al., 2022) stating that the implementation of good financial management practices can increase the profitability of a business. (Sinebe, 2023), stating that financial evaluation and planning show the least influence on business performance. (Srie Yuniawati et al., 2023) adding that the ability to evaluate financial statements ranging from forecasting to financial analysis, affects the profitability ratio as part of business performance. Flexibility in forecasting allows companies to respond to market changes more quickly and appropriately, which is essential for maintaining competitiveness (Grobler-Dębska et al., 2025). Formal adoption of predictive analytics is a must for companies that want to build resilience and competitiveness in today's fast-changing business competition (Akpe et al., 2021). Forecasting accuracy is one of the many possibilities of business analytics in the budgeting process that provides deep insights and better coordination in running a business (Bergmann et al., 2020).

Forecasting Theory explains that accurate business forecasting is heavily influenced by the quality and quantity of data, analysis methods and external factors such as seasonality and trends. This theory reinforces the relationship between variables such as budgeting, big data tools and competitiveness against accuracy forecasting. Forecasting theory is based on the premise that present and past knowledge can be used to make predictions about the future, particularly in analyzing historical values can be identified and applied effectively in the process of predicting future values (Petropoulos et al., 2022). Forecasting process enabling the modeling and optimization of future business activities. The application of forecasting tools in an integrated corporate governance system can make business activities more predictable, more manageable and more controllable (Yaremko et al., 2022).

This study aims to empirically examine the effect of forecasting accuracy on business performance. Specifically, the objectives of this research are to analyze how improvements in forecasting accuracy contribute to enhanced business performance, to evaluate the role of predictive technologies such as machine learning and data analytics in supporting forecasting effectiveness, and to provide practical references for companies in designing effective and efficient forecasting systems to support business decision-making. In addition, this study investigates whether human resource competence acts as a moderating variable that strengthens or weakens the relationship between forecasting accuracy and business performance.

Based on the literature review, several hypotheses are developed. The quality and quantity of data are fundamental in building reliable forecasting models. Clean, accurate, relevant, and sufficiently large datasets enable forecasting models to generate more precise predictions, while errors in historical data significantly reduce forecasting performance.

H1: Budgeting has a positive effect on forecasting accuracy.

The use of big data technologies such as analytics tools, machine learning, and cloud-based forecasting platforms enhances efficiency and predictive capability by accelerating the processing of large and complex datasets.

H2: Big data tools have a positive effect on forecasting accuracy.

Competitive pressure encourages organizations to improve planning accuracy and allocate greater attention to forecasting activities in operational and strategic planning.

H3: Competitiveness has a positive effect on forecasting accuracy.

Accurate forecasting enables companies to plan business activities more efficiently, reduce production and operational costs, improve customer service, and strengthen financial

performance. Therefore, forecasting accuracy is expected to contribute directly to improved business performance.

H4: Forecasting accuracy has a positive effect on business performance.

Furthermore, this study assumes that forecasting accuracy mediates the relationship between budgeting, big data tools, competitiveness, and business performance.

RESEARCH METHOD

This study employed a quantitative research approach aimed at testing the empirical relationships among variables. Forecasting accuracy is positioned as a mediating variable influenced by independent variables—budgeting, big data tools, and competitiveness while business performance serves as the dependent variable. The research design follows a causal framework, focusing on testing the influence of one variable on another.

Primary data were collected through a quantitative survey method, as it allows for more relevant and direct insights into companies' operational outcomes. Questionnaires were distributed to respondents in the hospitality industry, a sector that is highly dependent on demand forecasting to maintain operational efficiency, optimize production, and enhance overall business performance.

Purposive sampling was applied to determine the research sample. This technique was chosen based on specific criteria, including companies that have implemented forecasting systems for at least two years, possess measurable performance data, and are willing to participate in the study. This approach enables researchers to focus on the most relevant and informative respondents, thereby increasing internal validity, while also being effective under time and resource constraints that limit the application of probabilistic sampling.

Data analysis was conducted using SmartPLS version 4 with the Partial Least Squares–Structural Equation Modeling (PLS-SEM) technique. This method was selected due to its ability to test complex causal relationships with relatively small sample sizes and without strict assumptions of data normality. The minimum sample size follows practical quantitative research guidelines, suggesting 30–100 respondents to ensure estimation stability.

Data collection was carried out through electronic questionnaires distributed via professional networks, industry associations, and company referrals. Initial screening questions were included to ensure respondent eligibility. The questionnaire employed a five-point Likert scale ranging from strongly disagree (1) to strongly agree (5). Instrument development was based on prior literature and expert consultation. A pilot test involving 30 respondents was conducted to assess validity and reliability before full deployment. The final data collection period lasted two weeks, yielding 61 valid respondents.

RESULT AND DISCUSSION

The measurement model evaluation was conducted through outer loadings, convergent validity (Average Variance Extracted/AVE), construct reliability (Cronbach's Alpha and Composite Reliability), and discriminant validity using the HTMT criterion. The results indicate that most indicators demonstrate acceptable outer loading values, particularly for big data and forecasting accuracy constructs, which exhibit strong explanatory power.

All constructs achieved AVE values above the threshold of 0.50, confirming convergent validity. Discriminant validity was also established, as all HTMT values were below 0.90, indicating adequate conceptual distinction among constructs.

Reliability testing shows that Cronbach's Alpha and Composite Reliability values for all constructs exceed 0.70, demonstrating strong internal consistency and measurement stability. Multicollinearity assessment using the Variance Inflation Factor (VIF) indicates that most indicators fall below the recommended threshold of 3.0, although one big data indicator

exhibits a higher VIF, suggesting a potential multicollinearity issue that remains within tolerable limits for exploratory research.

Hypothesis Testing

Hypothesis testing was conducted to examine the significance of the relationships among variables. The analysis employed a bootstrapping technique, in which the data were processed to estimate path coefficients and their corresponding standard errors. The results are reported in the form of *t*-statistics and *p*-values. A hypothesized relationship is considered statistically significant when the *p*-value is less than the significance level of 0.05.

Table 1 presents the results of the hypothesis testing

Variable	Hypothesis Path	Coefficient	P-Value	Conclusion
Budgeting (X1)	Positive	0.264	0.046	Hypothesis supported
Big Data (X2)	Positive	0.450	0.004	Hypothesis supported
Competitiveness (X3)	Positive	0.032	0.375	The hypothesis is not supported
Forecast Accuracy (M)	Positive	0.555	0.000	Hypothesis supported
Adjusted R2:	0.296			
F-test:	8.46	sig. l0.000097		

Source: Processed by the researcher (2025)

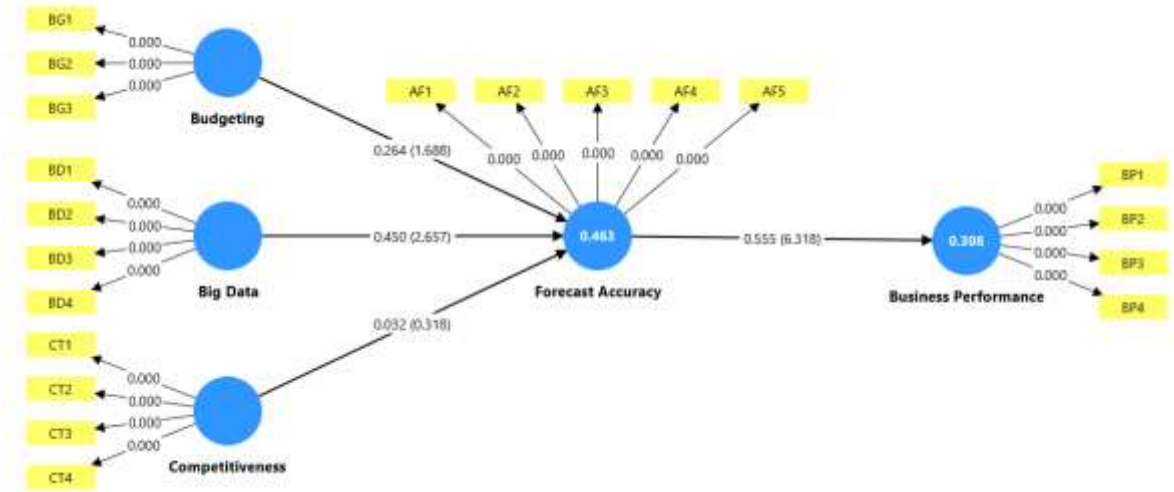


Figure 1. Research Model (SmartPLS)
Source: Processed by the researcher (2025)

The Influence of Budgeting on Forecast Accuracy

The hypothesis testing results show that budgeting has a positive and significant effect on forecast accuracy (β = 0.264; p = 0.046). Although the effect is moderate, the findings indicate that in a dynamic competitive environment, forecasting cannot rely solely on traditional planning mechanisms, but requires the support of data and information technology. The results suggest that a systematic and structured budgeting process, supported by adequate information, can help companies improve forecasting accuracy. Budgeting and forecasting can complement each other simultaneously to enhance operational performance (Bukh let lal., l2025). Conceptually, budgeting functions as a tool for coordination, control, and internal

communication, providing a baseline for planning and operational tolerance. These findings support Planning Theory, which emphasizes the importance of budgeting in organizational planning processes (Abbo, L2024).

The Influence of Big Data on Forecast Accuracy

The results show that big data has a positive and significant effect on forecast accuracy (β = 0.450; p = 0.004). The findings indicate that the higher a company's ability to collect, process, and analyze large and diverse datasets, the more accurate its forecasting process becomes. Theoretically, these results are consistent with Information Processing Theory, which emphasizes that organizations with greater information-processing capacity are better able to respond to uncertainty. Big data provides real-time and context-rich information that helps reduce forecasting errors. The findings are also in line with previous studies showing that the adoption of big data analytics improves predictive capability, demand planning, and operational decision accuracy (Mkhize let lal., l2025). Through machine learning, companies can further enhance their ability to predict trends and future business movements, thereby supporting strategic decision-making (Ganesan, L2024). Thus, big data plays an important role in improving forecast accuracy.

The Influence of Competitiveness on Forecast Accuracy

Unlike the previous two variables, the results show that competitiveness does not have a significant effect on forecast accuracy (β = 0.032; p = 0.375). In other words, the intensity of competition does not directly encourage companies to improve their forecasting performance. This finding provides an interesting theoretical contribution. Although market competition is often believed to encourage firms to enhance adaptability and responsiveness, the results of this study indicate that competitive pressure does not necessarily lead to improved forecasting. Several possible explanations can be considered: (a) companies may not yet have sufficient adaptive capacity to respond to competitive pressure through improved forecasting, and (b) competition may influence strategic decisions more strongly than operational practices such as forecasting. Empirically, the findings show that improvements in forecast performance are more influenced by internal capabilities (budgeting and big data) than by external factors such as competition.

The Influence of Forecast Accuracy on Business Performance

The results of this study indicate that forecast accuracy has a positive and significant effect on business performance (β = 0.555; p = 0.000), representing the strongest relationship in the model. These findings demonstrate that forecast accuracy directly improves business performance. Accurate forecasting enables organizations to enhance operational performance, even within a one-year planning period (Efat let lal., l2024). Forecast accuracy has strong predictive power for company performance and is closely related to profitability (Tanaka let lal., l2020). Accurate forecasting allows companies to: (a) improve operational efficiency, (b) optimize production capacity, (c) make more accurate strategic decisions, and (d) monitor operational costs more effectively. These results support the operations management, supply chain planning, and strategic management literature, which emphasizes forecasting accuracy as a key determinant of organizational performance.

CONCLUSION

This study concludes that big data and structured budgeting significantly enhance forecast accuracy, with big data being the most influential factor, while competitiveness does not exert a significant direct impact. Importantly, forecast accuracy itself strongly and positively affects business performance, underscoring its role as a critical mediator between organizational capabilities and performance outcomes. These findings highlight that internal factors data management and budgetary control are more pivotal than external competitive pressure in driving forecasting effectiveness. However, the study acknowledges limitations including a relatively small sample size, cross-sectional design, and reliance on self-reported data, which may affect generalizability and introduce perceptual biases. For future research, it is recommended to increase sample size, expand the range of industrial sectors, or apply longitudinal research designs to capture long-term dynamics among variables. Future studies may also include additional variables such as supply chain integration, technological capability, environmental uncertainty, or organizational learning as mediating or moderating variables. Furthermore, multigroup analysis can be applied to examine differences based on industry characteristics or organizational size. By incorporating additional variables and methodological approaches, future research is expected to provide a more comprehensive understanding of forecasting accuracy and business performance.

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